Bachelor Project: Chess Coach using Description and Defeasible Logic

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Outline

General Overview

Example 1: Baseline

Example 2: Database and Modal Logic

Validation

Recap

Main Idea

The main goal is to create a chess coach that focuses on **explainability** rather than **evaluation** of a given chess position.
3 types of logic are to be used:

- Description logic
- Defeasible Reasoning
- (optionally) Modal Logic

Pipeline Overview

1. User enters chess position (pgn)
2. Evaluation is done by outside source (api to stockfish or have it run locally, both possible)
3. Description logic is used to infer higher level concepts for the given position
4. Defeasible reasoning is used with the results of the DL inference (optionally: Modal logic in tandem)
5. Output is given to LLM
6. LLM generates explanation / answer

Example 1 input

User: Explain this position to me.

Figure 1: Carlsen vs Caruana, 2014 Springfield Cup

Example 1: Description Logic

Figure 2: Example of DL reasoning on a chess position

Example 1: DL output

After inference, we get a description of the current position. Classes such as:

- Under Attack
- Safe/Weak King
- Isolated Pawn
- Pawn chain
- Lines of sight (purple)
- Open File (Controlled / Open / Battle)
- Passive
- Controlled Squares

Can be used.

Example 1: Defeasible Reasoning

Now, with the description of the position we use Defeasible Reasoning, alongside the computer evaluation, to create an explanation of the position.
How? No idea.
Any advice is appreciated here, textbooks, articles, papers etc.
What I've read so far:
Article 1 Article 2
But, not nearly enough to feel confident in applying defeasible logic here.

Defeasible Reasoning

Regardless, the output would(could?) be along the lines of:

- Despite Black's king being weak, white has no way of punishing.
- Although white has a safe king, black has very strong lines of attack on white's king
- Although black has a "stupid" rook, it can be activated with an attack on white's rook by moving the bishop.
- Although white has an open file, it is being held by a singular, unsupported rook which can be attacked.

Example 1: Output

Now that we have a description of the position using DL, and have used defeasible logic for exception handling, we can generate an output with an LLM. For this example, a possible output could be:
Chess Coach: "Although black's king is under attack, white has no real way to capitalise on this weakness. A move like Bg4!!:

1. Attacks white's queen,
2. Opens up black's "stupid" rook on a8,
3. Attacks the sole defender of white's knight on h8,
4. ...

Although it may seem as if white is creating pressure on black's king, black is in time to counterattack! A tip to understanding these types of positions is to try to **explain in words** what your opponent can do to you.
Do you understand now why black is winning? What if black plays Qxg3, why is this a losing move for black?"

Example 2: How to use an Opening Database and/or Modal Logic

Given a position, an opening database could be utilised to find "ideas" within a position. (if similar descriptions, then look at goal of similar opening)

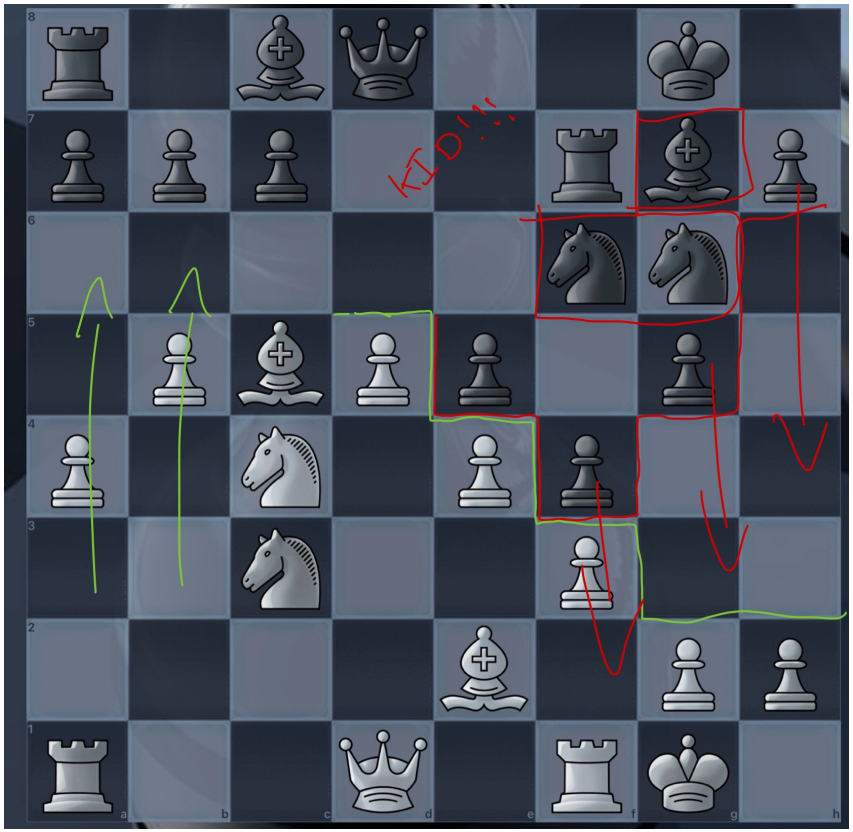

Figure 3: Gelfand vs Nakamura, World Team Championship 2010

Example 2: Modal Logic

Modal Logic could be used if the user asks a multi-move or planning type of question, examples of such questions:

- "What is blacks's plan forward in this position?"
  - For black to maintain an advantage, it is necessary for black to play h5. (Modal Logic)
  - Generally, in KID positions, black aims to expand on the kingside with f5 and g5. (Database)
- "If black moves the knight, is black still winning?"
  - It is possible for black to play Ng8 and still maintain an advantage.

Important note; using a database of openings might require a massive amount of upfront compute to put all the possible (or most common) openings through the DL reasoner **and** the defeasible reasoning portion, and maybe modal logic as well?

How do we Validate?

Currently, I see only three real options:

- Crowdsourcing (Reddit: r/chess, perhaps chess.com forum)
- Chess books. Chess books usually contain human centred explanations. Upload same position and check if semantically similar.
- Local Chess clubs.

Quick Recap

- **Explainability** not Evaluation
- Overview of the Pipeline:
  1. Input
  2. DL to describe position
  3. Defeasible reasoning to handle exceptions
  4. Throw everything into an LLM
  5. LLM generates output
- By far the weakest point is Defeasible reasoning (help)